
1. Project Problem Statement

We are building a **personal AI that runs locally**, learns routines and preferences, plans tasks, uses tools, and protects privacy. This is the **full problem** that all models must solve.

2. Baseline Model

Baseline: Open Jarvis [published research paper] We reproduce one configuration of Open Jarvis as our baseline. It is a **complete personal-AI agent** that includes:

- Planning
- Memory
- Tool use
- Privacy rules
- Local execution
- Cost + latency measurement

This baseline is our reference point for comparison.

3. Our Four Models [Each is a Full Agent]

Each model solves the **same full project problem** but uses a **different core method**.

Model 1 — Planning + RAG Agent

A full personal AI agent whose main technique is hierarchical planning + retrieval.

Model 2 — Rou5ne & Preference Learning Agent

A full personal AI agent that learns user habits and preferences over time. **Model**

3 — Contextual Bandit Agent

A full personal AI agent that chooses the best action or suggestion using a bandit policy. **Model**

4 — Safety & Privacy Cri5c Agent

A full personal AI agent that predicts risk and blocks unsafe or privacy-violating actions.

4. Unified Benchmark [Same Tasks for All Models]

All 5 agents [baseline + 4 models] are tested on the **same tasks**, such as:

- Summarizing messages
- Checking calendar conflicts
- Extracting tasks from emails
- Planning a day
- Retrieving files
- Applying privacy rules
- Detecting unsafe actions
- Remembering preferences

- Every agent receives the **same input**.

5. Unified Metrics [Same Scoreboard for All Models]

We evaluate all agents using the same metrics:

- Task success
- Planning accuracy
- Memory accuracy
- Safety violations
- Privacy violations
- Latency
- Cost per user-day • Intervention rate
- Preference regret

This ensures fair comparison.

6. Comparison Table [Example Format]

Metric	Baseline	Model 1	Model 2	Model 3	Model 4
Task Success	68%	74%	71%	77%	81%
Planning Accuracy	62%	71%	65%	72%	68%
Memory Accuracy	58%	66%	72%	61%	64%
Safety Violations	4	2	3	2	1
Privacy Violations	3	1	2	1	1
Latency	42tims	18tims	25tims	21tims	31tims

This is the structure your team will use.

7. Short Summary

Baseline = Open Jarvis full agent

Model 1 = full agent [planning + RAG]

Model 3 = full agent [bandit policy].

All 5 solve the same full problem

Model 2 = full agent [routine learning]

Model 4 = full agent [safety critic]

All 5 use the same tasks
All 5 use the same metrics
All 5 are compared in one table.

